

PLASTIC UPCYCLING

Integrated low-temperature PVC and polyolefin upgrading

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Polyolefins and their chlorinated derivatives such as polyvinyl chloride (PVC) are among the most prevalent plastics in global production and waste streams. Traditional waste-to-energy methods such as incineration and pyrolysis, as well as most chemical upcycling methods for PVC utilization, require thorough, high-temperature dechlorination to prevent the release of toxic chlorinated compounds. We present here a strategy for upgrading discarded PVC into chlorine-free fuel range hydrocarbons and hydrogen chloride in a single-stage process catalyzed by chloroaluminate ionic liquids. This approach offsets endothermic dechlorination and carbon-carbon bond cleavage with exothermic alkylation and hydrogen transfer by isobutane or isopentane in a low-temperature tandem process. The light isoalkanes are available from refinery processes and partly from recycling of the product stream. This process is suitable for handling real-world mixed and contaminated PVC and polyolefin waste streams.

Polyolefins (polyethylene and polypropylene) represent the largest fraction of global plastic production (~50%), and their chlorinated derivative polyvinyl chloride (PVC) accounts for an additional ~10%. This combined class dominates plastic waste streams because of its resistance to degradation (1, 2). PVC, with its high Cl content [~57 weight % (wt %)], poses more severe environmental issues than other polymers when disposed of. The physical separation and the mechanical recycling of mixed waste face technical challenges and thus are typically economically uncompetitive (fig. S1) (3, 4). Traditional waste-to-energy methods such as incineration necessitate thorough PVC dechlorination to prevent the release of toxic chlorinated compounds such as dioxins and polychlorinated biphenyls (5). Moreover, PVC's flame-retardant properties hinder energy recovery efficiency (6), leading to a substantial portion of it ending up in landfills (7). Recognizing these challenges, the European Chemicals Agency (ECHA) and the US Environmental Protection Agency (EPA) in 2023 initiated reviews on regulatory actions for PVC and related substances (8, 9), underscoring the urgency of addressing their environmental impact (10).

Chemical upcycling of PVC and mixed-waste streams offers a promising approach to producing a range of tailored chemicals, intermediates, fuels, lubricants, and new polymers for existing markets (11–14). These procedures are initiated by dechlorination using various Cl scavengers [alkaline solutions (15, 16), basic ionic liquids (17–19), and solids such as CaCO₃ (20), CaO (21), ZnO (22, 23), Al₂O₃ (21), Fe₃O₄ (24), and Mg–Al oxides (25)], followed by C–C bond cleavage of the remaining hydrocarbon backbones (26, 27). The latter step resembles

the deconstruction of polyolefins and polyenes (28, 29), exemplified by hydrogenolysis and/or hydrocracking on supported metal nanoparticles (15). Although the dechlorination step prevents catalytic active sites such as commonly used reduced metals from being poisoned by HCl or Cl-containing compounds, it leads to complex operation, reduced efficiency, and increased energy consumption (28).

Substantial progress has been made in recent years in developing catalytic transformations of polyolefins under mild conditions with tandem processes that couple endothermic C–C bond cleavage with exothermic reactions (30, 31). The selected reactions use a variety of co-reactants, including H₂, light paraffins, olefins, and O₂. These reactions fall under various categories, such as hydrogenolysis (32, 33), hydrocracking (34–36), alkylation (37–39), metathesis (40–42), cyclization (43, 44), and oxidation (45, 46). The strategies conceptually and practically minimize energy requirements and optimize conversion thermodynamics, enabling complete polymer conversion at mild conditions (47).

In principle, endothermic dechlorination could also be coupled within the process, enabling a similar approach to converting PVC. However, the lower H/C ratio in PVC compared with polyolefins requires substantial H addition or redistribution in addition to forming C–C bonds. The thermodynamic calculations indicate that dehydrochlorination and C–C cleavage alone are restricted by equilibrium constraints at low temperatures (Fig. 1B). We previously found that dehydrochlorination can be kinetically feasible at near ambient temperatures when catalyzed by anhydrous AlCl₃ or AlCl₃-based ionic liquids through chloride-transfer mechanisms (38, 39). Very recently, Gao *et al.* reported a room-temperature co-upcycling approach to processing PVC and polypropylene (PP), achieving very high yields of liquid hydrocarbons without any additional H donor (48). However, beyond overcoming the thermodynamic challenges of endothermic dechlorination and C–C bond cleavage, the inherently low H/C ratio in PVC and PP necessitates substantial H input (49). Without this, the resulting alkenes undergo oligomerization or participate in intermolecular hydride transfer, generating unsaturated carbenium ions that rapidly cyclize in subsequent steps (37). Therefore, the reported product distribution, particularly the high fraction of alkenes, raises questions of thermodynamic feasibility. The oligomerization of alkenes is exothermic and occurs significantly faster than alkylation, leading readily to the formation of acid-soluble conjugate polymers, commonly referred to as “acid-soluble oil (ASO) or red oil,” which deactivate the chloroaluminate ionic liquid catalysts (50, 51). This phenomenon explains why industrial C₄ alkylation typically requires large excesses of isobutane (iC₄) relative to alkenes to minimize ASO formation (52).

Conceptually, coupling the endothermic dechlorination and C–C bond breaking and the exothermic alkylation, including the H redistribution, should suffice to make the combined reaction sufficiently exergonic to allow full conversion (Fig. 1). The light isoalkanes iC₄ and isopentane (iC₅) required are readily available as side products from processes such as fluid catalytic cracking, hydrocracking, and paraffin isomerization (53). Typically, these light isoalkanes are upgraded by alkylation with C₃–C₅ olefins, producing highly branched, gasoline-range alkanes (52, 54). Coprocessing PVC, polyolefins, and light isoalkanes could ideally serve as a potential feedstock for alkylation in existing refineries.

Building on this, we report the use of AlCl₃-based ionic liquids for the low-temperature catalytic conversion of PVC and polyolefin waste into liquid hydrocarbons and HCl (Fig. 1A). The process couples PVC dechlorination and C–C bond cleavage with the alkylation of iC₄ or iC₅ into a tandem operation. It ensures the preservation of all C atoms in the Cl-free end products, aligning with refinery operations for efficient joint upcycling of PVC, polyolefins, and isoalkanes. Converting PVC into nonchlorinated hydrocarbons prevents the reintroduction of highly toxic and carcinogenic organics. Conversely, the recovered HCl can be safely neutralized or reused in diverse applications in modern industries. For example, it serves as a raw material in vinyl chloride

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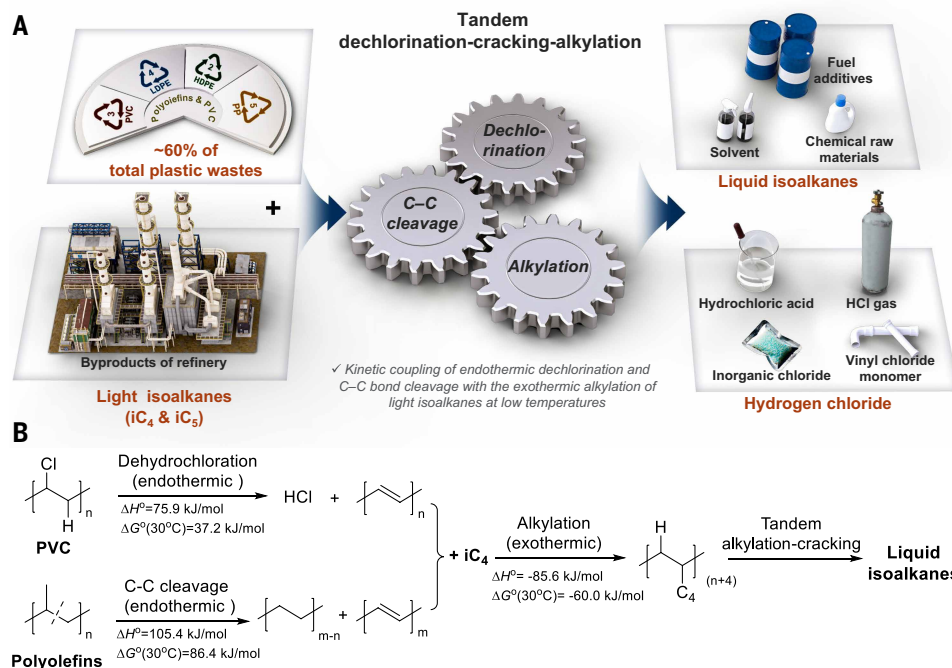


Fig. 1. Proposed co-upcycling of PVC, polyolefins, and light isoalkanes. (A) Schematic illustration depicting the coupling of mixed PVC and polyolefin waste plastics with light isoalkanes through a tandem dechlorination, alkylation, and cracking process, producing liquid isoalkanes and HCl in a single stage. (B) Proposed sequence of key reaction steps integrating endothermic dehydrochlorination and C–C bond cleavage with exothermic alkylation using iC_4 , with the respective Gibbs free energy (ΔG°) and enthalpy (ΔH°) shown. The corresponding ΔG and ΔH values were calculated using HSC Chemistry 10 (61). See details in fig. S2 and table S1.

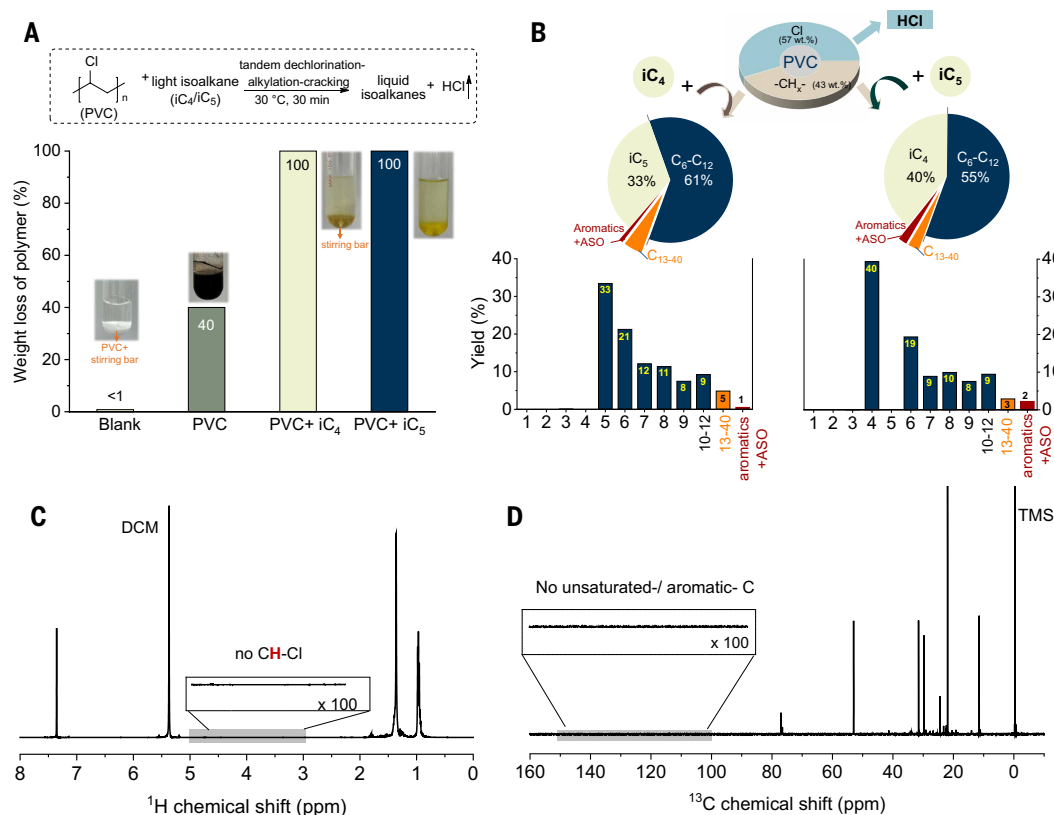


Fig. 2. Tandem dechlorination, alkylation, and cracking of PVC at room temperature. (A) One-pot catalytic upcycling of PVC and isoalkane into liquid alkanes over $AlCl_3$ -based chloroaluminate ionic liquids compared with isoalkane-free PVC deconstruction. Reaction conditions were as follows: PVC, 200 mg; iC_4/iC_5 , 800 mg; catalyst, 1 mmol; DCM, 3 ml; temperature, 30°C for 30 min. The insets show snapshots taken after the reactions and the addition of extraction agents. (B) Product distribution from PVC reaction with iC_4 and iC_5 . (C and D) 1H (C) and ^{13}C NMR (D) spectra of products in the upper organic layer after PVC/ iC_5 reaction, confirming that the organic products are Cl-free alkanes.

monomer production in new PVC synthesis and as a chemical precursor in metal processing, water treatment, food production, pharmaceuticals, and the petroleum industry (55). Currently, HCl is primarily produced through several energy-intensive processes, such as direct synthesis from H and Cl at temperatures exceeding 400°C, the HCl process involving concentrated H₂SO₄ and NaCl, and the recovery of by-products from organic chlorination reactions (56).

Reaction development

For our initial studies, we chose commercial PVC [weight-averaged molecular weight (M_w), ~62,000; number-averaged molecular weight (M_n), ~35,000] and isoalkanes (iC₄ or iC₅). In a glass tube reactor, 200 mg of PVC was combined with 800 mg of either iC₄ or iC₅, 3 ml of dichloromethane (DCM), and 1 mmol of AlCl₃-based butylpyridinium chloroaluminate ionic liquid catalyst (denoted as C₄PyCl-2AlCl₃) at 30°C under ambient pressure (Fig. 2A). Control experiments without catalyst confirmed <1% weight loss of PVC, indicating negligible background activity. By adding 1 mmol of the catalyst, the reactions of PVC with either iC₄ or iC₅ or their mixtures achieved 100% polymer conversion in just 30 min (see details in fig. S3). By contrast, control experiments without light isoalkanes yielded only 40% polymer weight loss even when the reaction time was extended to 4 hours under identical conditions, with black solid residues observed (Fig. 2A). Characterization of the black residues revealed that the solids were mainly composed of polyenes (fig. S4). Elemental analysis and energy-dispersive x-ray spectroscopy showed that the Cl content was ~23 wt %, much lower than in pure PVC (~54 to 57 wt %), indicating that the weight loss was mainly caused by dehydrochlorination (figs. S5 and S6 and table S2) rather than by polymer chain cleavage.

After the complete conversion of PVC by processing with light isoalkanes, we examined the distributions of C and Cl in products (see details in figs. S7 and S8; the results are summarized in Table 1). The results from PVC and iC₅ processing, as illustrated in Fig. 2B, indicated that ~55 wt % of the products (C₆-C₁₂ isoalkanes) fell within the gasoline range, whereas 40 wt % was iC₄, as confirmed by gas chromatography-mass spectrometry (fig. S9). Table S3 summarizes the product distribution of all C₈ products, showing that nC₈ accounted for only 5.5%, whereas the remaining products were branched C₈ alkanes. The ¹H (Fig. 2C) and ¹³C nuclear magnetic resonance (NMR) (Fig. 2D) spectra further confirmed that the products in the upper layer were saturated Cl-free hydrocarbons. The lower ionic liquid layer exhibited a hydrocarbon composition of 2 wt % unsaturated compounds (aromatics and ASO). The two-dimensional ¹H-¹³C NMR spectra identified aromatic signals only in this layer (fig. S10, A and B), in agreement with the gas chromatography-mass spectrometry findings (fig. S10C). Similarly, processing PVC with iC₄ gave 94 wt % isoalkanes, consisting of 33 wt % iC₅ and 61 wt % C₆-C₁₂ (Fig. 2B and figs. S11 and S12). The headspace in both reactions primarily contained iC₄ and iC₅, with negligible propane (C₃), n-butane (nC₄), and n-pentane (nC₅) (< 0.1 wt %), and was free from methane (CH₄), ethane (C₂), and unsaturated light hydrocarbons

(figs. S13 and S14). These findings are consistent with our previous hypothesis on polyolefin upcycling that the reaction proceeds through a carbenium ion mechanism, with C-C bond cleavage predominantly occurring through type A β-scission (fig. S15) (37).

With either iC₄ or iC₅ as the sole reactant, <0.5 wt % conversion was observed (37). iC₅ disproportionation did occur in the presence of carbenium ions [e.g., by adding a significant amount of tert-butyl chloride (TBC) as a co-reactant], predominantly producing iC₄ and iC₆ (fig. S16) (57). Control experiments using iC₄ as a co-reactant showed that it was far superior to iC₅ (see details in Fig. 2B), effectively suppressing the side disproportionation reaction. However, iC₅ disproportionation did not interfere with the reaction, because these light isoalkanes serve as alkylating agents in the process.

The experimental stoichiometric mass ratios of converted PVC to iC₄ and iC₅ were ~1:2.5 and ~1:3 (Table 1), respectively, which aligns well with the theoretical expectations (see details in fig. S17). Control experiments demonstrated that when the PVC to iC₅ ratio was set as 1:2, full PVC conversion was achieved; however, this ratio led to the formation of ~21 wt % unsaturated compounds (see details in table S4 and fig. S18). Specifically, in the coprocessing of iC₄ and iC₅ mixtures, the stoichiometric mass ratio of converted PVC to mixed isoalkanes decreased to 1:2.4, resulting in 95 wt % liquid isoalkanes ranging from C₆ to C₁₂ (fig. S19). Therefore, this tandem approach effectively couples endothermic dechlorination and C-C bond cleavage with exothermic alkylation of light isoalkanes (fig. S20), ensuring both kinetic and thermodynamic efficiency.

We assessed the Cl mass balance of PVC and the Cl distribution within products and catalysts. Most of the Cl in PVC was converted to HCl (~61%, Table 1), with the remaining chemically bound as AlCl₄⁻ anions within the ionic liquid (58) (~35% as derived from Raman spectroscopy analysis; fig. S21). The continuous addition of PVC, with complete conversion, did not lead to a notable increase in the concentration of AlCl₄⁻ ions, whereas the yield of released HCl reached as high as ~80% (see details in table S5).

Mechanistic studies

Having established the thermodynamic boundary conditions and the principal kinetic feasibility to convert PVC in tandem with alkylation, we next explored the reaction pathways involved. Attenuated total reflection-infrared (ATR-IR) spectroscopic analysis of the solid residues at various reaction times during the PVC conversion in the absence of an isoalkane (Fig. 3A) showed at first characteristic PVC vibrations such as C-Cl stretching (605 cm⁻¹ and 632 cm⁻¹), (Cl)C-H bending (1250 cm⁻¹), and C-H bending (1330 cm⁻¹), gradually diminishing in intensity with reaction time. The decrease of these bands was accompanied by the emergence of new bands characteristic of C=C stretching vibrations of unsaturated bonds in the spectral regions of 1642 cm⁻¹ and 1605 cm⁻¹. These results strongly support the dechlorination pathway (59, 60).

A markedly different kinetic trajectory of PVC was observed when using iC₅ as a co-reactant. The spectra of the solid residue were nearly

Table 1. Summary of one-pot catalytic PVC/isoalkane upcycling into liquid alkanes over AlCl₃-based chloroaluminate ionic liquids. Reaction conditions were as follows: PVC, 200 mg; iC₄ or iC₅, 800 mg, light paraffin mixture (C₄/C₅: ~1:1) C₄PyCl-2AlCl₃, 1 mmol, DCM, 3 ml; and temperature 30°C for 30 min.

Entry	Stoichiometric mass ratio	Hydrocarbon distribution (wt %)*						Cl (wt %)			
		C ₃	C ₄	C ₅	C ₆ -C ₁₂	C ₁₃ -C ₄₀	Olefins / aromatics	C balance (%)	HCl (wt %)	Cl in ionic liquid (wt %) [†]	Cl balance (%)
PVC/iC ₄	1 / 2.5	< 0.1	-	33	61	5	1	96	61	35	96
PVC/iC ₅	1 / 3	< 0.1	39	-	55	3	2	96	60	35	95
PVC/iC ₄ /iC ₅	1/1.1/1.3	< 0.1	-	-	95	4	1	93	55	35	90

*Methane, ethane, ethene, and propene were not formed.

[†]These values were determined by Raman spectroscopy.

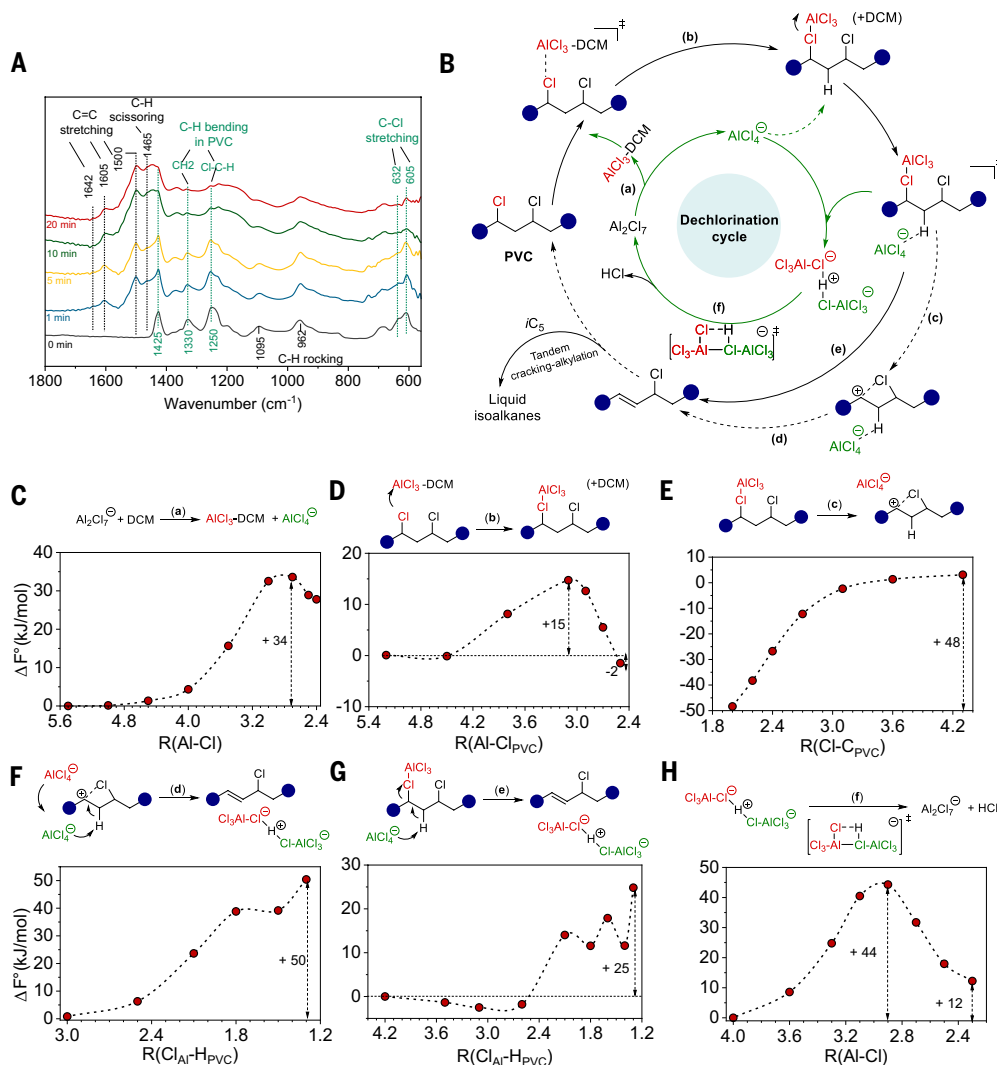


Fig. 3. Dehydrochlorination of PVC. (A) ATR-IR study of PVC deconstruction into alkanes with anhydrous AlCl_3 catalyst in the absence of isoalkane as co-reactant. Reaction conditions were as follows: PVC, 200 mg; $\text{C}_4\text{PyCl-2AlCl}_3$, 1 mmol; temperature, 30°C . The PVC residues were collected at intervals of 1.5, 10, and 20 min. (B to H) Proposed reaction mechanism for the dehydrochlorination of PVC to form polyene and HCl. (B) Illustration of chloroaluminate species driving the dehydrochlorination of PVC into polyene and HCl through carbenium ion chemistry, further catalyzing the coupling of alkylation and cracking reactions to produce liquid alkanes. (C to H) Reaction free energies calculated from AIMD trajectories. The Helmholtz free energy (ΔF°) as a function of internuclear distance calculated using an AIMD-based Blue Moon ensemble approach, is plotted. Dashed curves represent a fit of the data, serving as a guide for the eye. (C) Reaction of Al_2Cl_7^- and DCM to produce an AlCl_3 -DCM adduct. (D) Formation of the AlCl_3 -PVC adduct. (E) Barrier for dissociation of AlCl_4^- from PVC. (F) Deprotonation of a partially dechlorinated PVC by AlCl_4^- . (G) Simultaneous PVC deprotonation by AlCl_4^- and dechlorination by AlCl_3 . (H) Recombination of AlCl_4^- - H^+ - AlCl_4^- , produced in both (E) and (G), to form Al_2Cl_7^- and HCl. For detailed structures, see figs. S26 to S32.

identical to those of pure PVC (fig. S22), which suggests that the dechlorination, the cracking of PVC, and the alkylation with iC_5/iC_4 occurred concurrently. To avoid spectroscopic interference from the presence of the solvent DCM and butylpyridinium cations in the ionic liquid (fig. S23), we directly monitored the reaction of PVC using iC_5 as a co-reactant and AlCl_3 as the catalyst at 30°C (fig. S24). We found that the intensities of the iC_5 bands decreased significantly within 5 min (fig. S25). The bands characteristic of C=C double bonds were absent in reactions with iC_5 . This implies that the reaction started with Cl removal by dehydrochlorination, followed by rapid alkylation with iC_5 . The saturated, dechlorinated, and alkyl-functionalized C backbone would then subsequently crack and the olefinic products would react with the isoalkyl carbenium ions to produce isoalkane fragments after

hydride transfer from other iC_4 or iC_5 molecules. The IR spectral evolution strongly suggests that these steps occurred in a concerted manner.

To study the probability of reaction steps, we used ab initio molecular dynamics (AIMD) simulations coupled with a Blue Moon ensemble method to gain insights into the dehydrochlorination process, as shown in Fig. 3B and figs. S26 to S32. We hypothesized that the reaction begins in either of two ways. In the first, PVC reacts with AlCl_3 , which is generated in situ from the DCM solvent-mediated dissociation of Al_2Cl_7^- to form AlCl_4^- and an AlCl_3 -DCM adduct (39) (Fig. 3C and fig. S26) with a free energy barrier ΔF° of 34 kJ/mol. After this, the chloride transfer between AlCl_3 -DCM and Cl-PVC to form a PVC-derived AlCl_3 - Cl_{PVC} complex presents a barrier as low as 15 kJ/mol (Fig. 3D and fig. S27). The second option is that PVC may directly dissociate Al_2Cl_7^- to form AlCl_4^- and the AlCl_3 - Cl_{PVC} complex ($\Delta F^\circ = 25$ kJ/mol fig. S28). AlCl_3 - Cl_{PVC} is a relatively stable bond, with calculations suggesting that it is closer to an adduct, primarily retaining the C-Cl covalent bond character rather than a Cn^+ and AlCl_4^- ion pair. Once the AlCl_3 - Cl_{PVC} complex forms, dehydrochlorination then occurs through two potential mechanisms: either AlCl_3 dechlorinates PVC to form AlCl_4^- and a carbocation stabilized in the form of a halonium ion (Fig. 3E and fig. S29, $\Delta F^\circ = 48$ kJ/mol), followed by deprotonation of an adjacent C to produce an alkene (Fig. 3F and fig. S30, $\Delta F^\circ = 50$ kJ/mol), or a concerted dehydrochlorination may occur after the formation of the AlCl_3 - Cl_{PVC} . Then, if a second AlCl_4^- is available to deprotonate the adjacent C, simultaneously forming an alkene and a protonated AlCl_4^- ; this concerted process significantly lowers the activation barrier (Fig. 3G and fig. S31, $\Delta F^\circ = 25$ kJ/mol). During dehydrochlorination,

an AlCl_4^- - H^+ - AlCl_4^- complex forms, wherein the proton is stabilized between two AlCl_4^- . This dual stabilization is essential as a single AlCl_4^- lacks sufficient basicity to abstract the H^+ . Although H^+ - AlCl_4^- remains unstable, the AlCl_4^- - H^+ - AlCl_4^- complex is a stable intermediate that facilitates the H^+ abstraction. Our NVT (number of particles, volume, and temperature) trajectories confirmed the stability of this intermediate, which recombined (Fig. 3H and fig. S32, $\Delta F^\circ = 44$ kJ/mol) to produce HCl and Al_2Cl_7^- , which can subsequently reenter the dechlorination cycle. Therefore, we conclude that the initial dehydrochlorination step does not require isoalkanes as co-reactants, leading to the production of polyenes. To produce saturated alkanes, H-rich iC_4/iC_5 are required to facilitate the breaking of C-C bonds to transfer H and to form C-C bonds by alkylation.

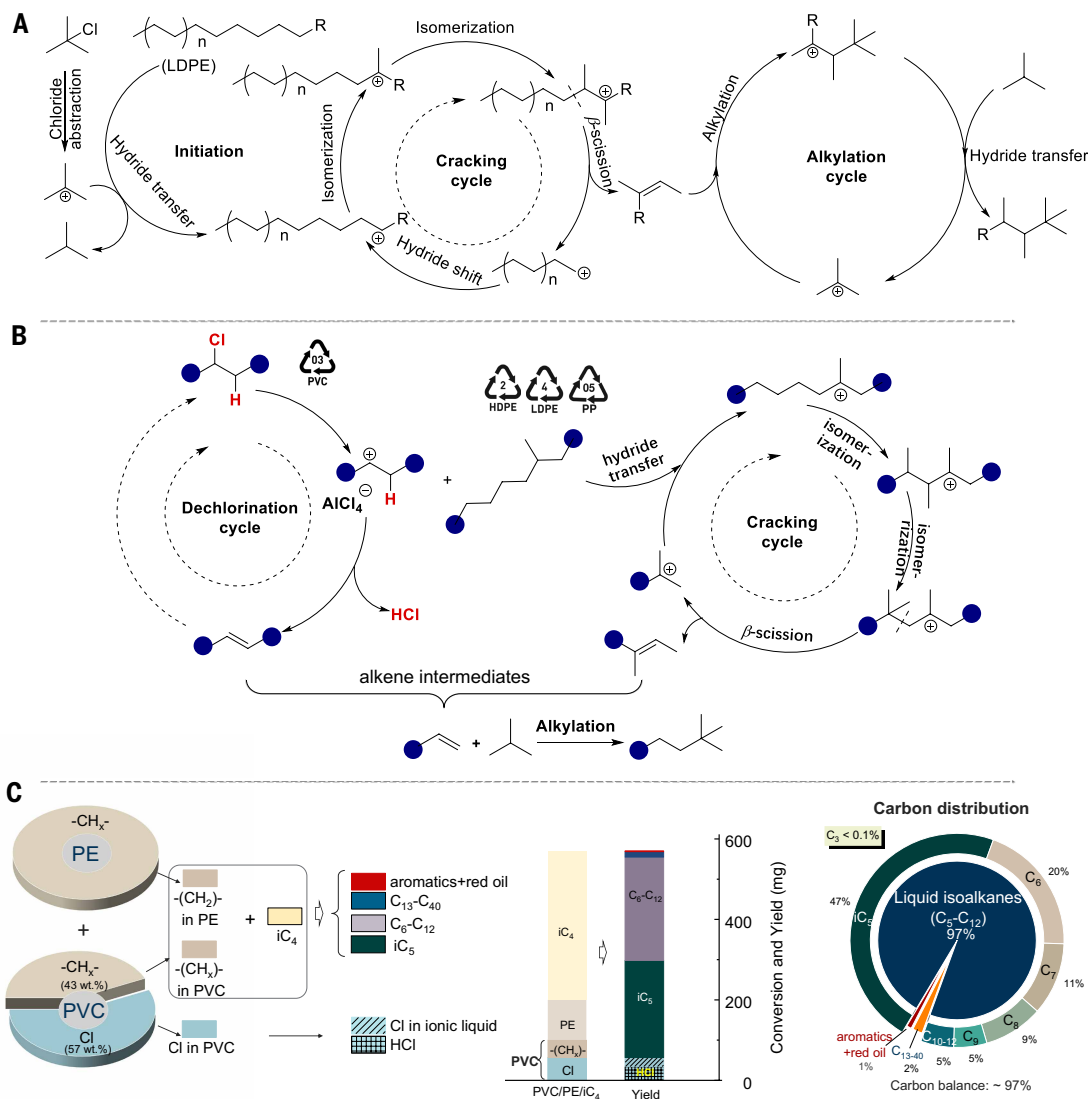


Fig. 4. Co-upcycling of mixed PVC and polyolefins with iC_4 . (A) Reaction mechanism for the tandem cracking and alkylation process of a polyolefin with iC_4 . [Figure used with permission from (37). Copyright 2023, AAAS]. (B) Proposed reaction mechanism for co-upcycling of mixed PVC and polyolefins with light isoalkanes through tandem dehydrochlorination, cracking, and alkylation. (C) Conversion and yield of coprocessing of mixed PE, PVC, and iC_4 , with an experimental stoichiometric mass ratio of 1/1/3.7, producing alkanes and HCl and the corresponding C distribution. Reaction conditions were as follows: commercial PE ($M_w \sim 4000$, $M_n \sim 1700$), 0.1 g; PVC, 0.1 g; iC_4 , 1.0 g; $C_4PyCl-2AlCl_3$, 1 mmol; DCM, 7 ml; temperature, 70°C for 30 min.

Coprocessing of PVC with polyolefins and postconsumer waste applications

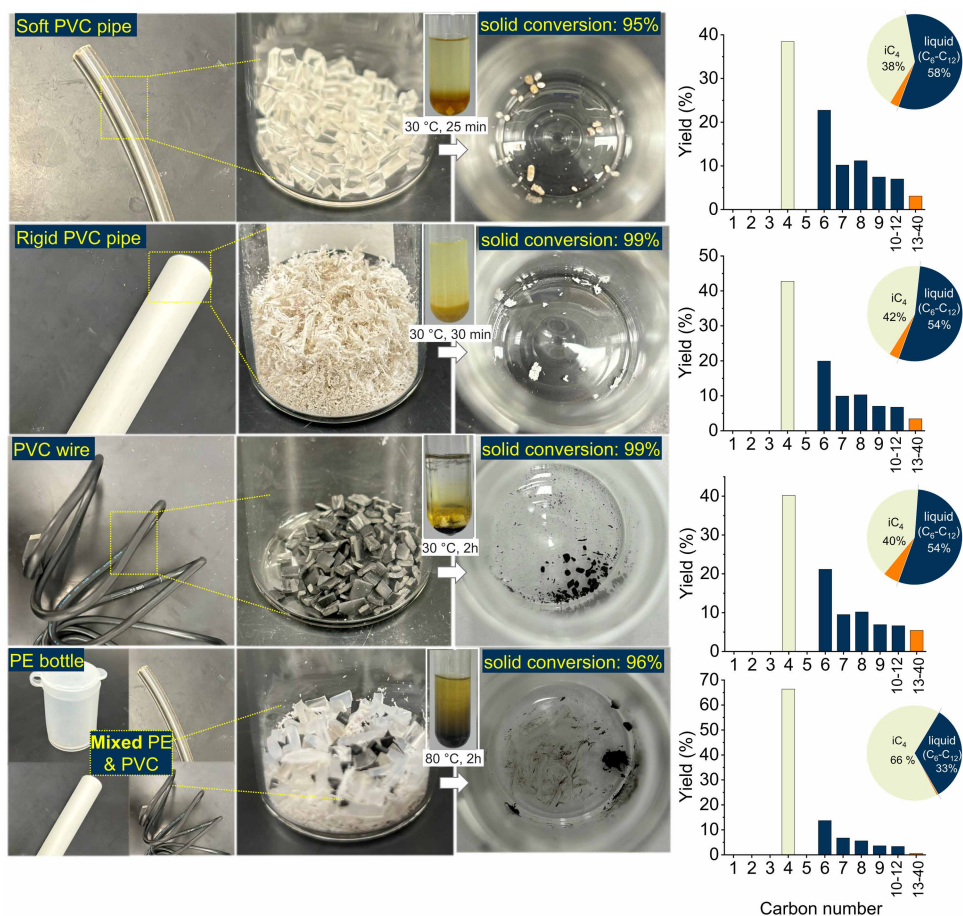
The tandem process achieves a complete catalytic transformation of mixed PVC and polyolefins in a single stage, substantially expanding and enhancing the thus-far reported conversion of polyolefins alone (fig. S33) (37). The process, involving polyolefin and isoalkane, typically necessitates small amounts of TBC as a carbenium ion initiator to avoid low reaction rates and complex initiation processes (Fig. 4A). PVC can be regarded as a source of carbenium ions because of its structural resemblance to long chains of interconnected TBC monomer units. Therefore, coprocessing PVC with polyolefins notably improves conversion rates of polyolefin deconstruction without the initiator (fig. S33). The carbenium ions generated from PVC dechlorination appeared to activate polyolefin chains through successive hydride transfers at low temperatures, eliminating the need for the TBC initiator (Fig. 4B). We posit that an increased fraction of PVC enhances carbenium ion production through dechlorination, which in turn

accelerates reaction rates. For instance, with a PVC to polyethylene (PE) ratio of 1:1 (as shown in Fig. 4C, with a stoichiometric mass ratio of $PVC/PE/iC_4 = 1/1/3.7$), the mixed polymer achieves full conversion within 30 min, yielding ~97 wt % liquid isoalkanes (C_5-C_{12}).

The current studies specifically focus on low-temperature catalytic transformations conducted in a single-stage process, deliberately sidelining the complexities introduced by feedstock purity to primarily focus on proof-of-concept demonstrations. The applicability of the catalytic tandem process to actual waste products is demonstrated by converting a broad range of PVC postconsumer waste products to alkanes at room temperature and ambient pressure (Fig. 5). PVC postconsumer waste, including various soft and rigid PVC pipes and wires, exhibited comparable reactivity at 30°C. The complexity of mixed real-world polymers may introduce various additives and impurities to the ionic liquid. The potential specific impact of these components must be addressed separately. Although additives may influence reaction rates, impurities are expected to predominantly accumulate in the denser

Fig. 5. Room-temperature deconstruction of postconsumer PVC waste into liquid alkanes.

Product distribution of tandem cracking and alkylation of postconsumer polyolefin waste with iC_5 with $C_4PyCl-2AlCl_3$. Note that no methane, ethane, or propane was detected; the products consisted exclusively of branched alkanes. Reaction conditions were as follows: postconsumer PVC waste, 200 mg; iC_5 , 800 mg; $C_4PyCl-2AlCl_3$, 2 mmol; and DCM, 3 ml. Reaction temperatures and times were as follows: soft PVC pipe, 30 °C for 25 min; rigid PVC pipe, 30 °C for 30 min; PVC wires, 30 °C for 2 hours; and a mixed-waste batch comprising 50 mg each of soft and rigid PVC pipes, 50 mg of PVC wires, and 150 mg of PE bottles, 80 °C for 2 hours. The insets show snapshots taken after the reactions and the addition of extraction agents.



ionic liquid phase. By contrast, the organic phase yielded significant amounts of Cl-free liquid alkanes (fig. S34) and gaseous iC_4 within 2 hours. The corresponding product distributions are illustrated in Fig. 5. Mixing PVC postconsumer waste with rigid high-density PE bottles also resulted in nearly quantitative solid conversion at 80 °C, predominantly producing iC_4 . The iC_5 disproportionation inevitably occurred in the presence of carbenium ions, but was limited largely to the initial stages of the reaction. Although ~40 to 66% of the product consists of gaseous iC_4 , it can be recycled as alkylating component along with unreacted iC_5 , highlighting the efficiency and sustainability of the proposed process.

Our strategy addresses the crucial issue of product distribution in the conversion of mixed plastic waste to fuel and is unaffected by the heterogeneity of the plastic mixture in the feed stream or impurities that could modify or poison catalysts. DCM dilutes the ionic liquid, lowering the viscosity and the local mass-transfer barriers, and is fully recycled through the top of the distillation product slate, including all light products.

Overall, this work opens a transformative and scalable approach for upcycling mixed PVC and polyolefin waste in a single-stage process that enhances efficiency and cost-effectiveness by reducing energy use, equipment needs, and intermediate steps, leading to significant savings and less environmental impact. The method supports a circular economy by converting diverse plastic waste into valuable products in a single step, improving product quality and facilitating scalability for industrial applications.

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SUPPLEMENTARY MATERIALS

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Integrated low-temperature PVC and polyolefin upgrading

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Editor's summary

Polyvinyl chloride (PVC) is a widely used plastic, particularly in plumbing applications, but its end-of-life processing can produce toxic chlorocarbon compounds, a growing problem as waste continues to accumulate. Zhang *et al.* report an efficient protocol to remove the chloride in PVC waste as hydrogen chloride using a room-temperature chloroaluminate ionic liquid medium. Mixed-waste streams including polyethylene are also compatible with this chemical process. The carbon chains are concurrently broken down into fuel-range hydrocarbons. —Jake S. Yeston

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